

E-Commerce Sales Data Analysis

Power BI Assignment 1

June 2026

Data Import | Transformation | Merging | Aggregation | Modeling

Project Objective

As a Data Analyst, this project involves working with an e-commerce sales dataset to perform data preparation and modeling using Power BI and Power Query Editor. The dataset contains multiple related files representing order-level, product-level, and sales target information. The focus is on data import, transformation, cleansing, aggregation, and relationship modelling while ensuring consistency, accuracy, and business relevance.

Dataset Description

Three CSV datasets were used in this project:

Dataset	Key Columns
List of Orders	Order ID, Order Date, Customer Name, City, State, Amount, Profit
Order Details	Order ID, Category, Sub-Category, Quantity, Amount, Profit
Sales Target	Category, Month of Order Date, Target

Column Description

The table below lists all columns across the three datasets, including derived and calculated columns.

Column Name	Table	Description
Order ID	List of Orders / Order Details	Unique identifier for each order
Order Date	List of Orders	Date when the order was placed
CustomerName	List of Orders	Full name of the customer (formatted to Proper Case)
City	List of Orders	City where the order was placed
State	List of Orders	State where the order was placed
Amount	List of Orders / Order Details	Total monetary value of the order (Fixed Decimal)
Profit	List of Orders / Order Details	Profit earned from the order
Location	List of Orders	Derived column: City and State merged as City, State
Profit Margin	List of Orders	Calculated column: Profit divided by Amount
Profit Status	List of Orders	Conditional column: Loss, Break-Even, or Profit
Category	Order Details / Sales Target	Product category (Furniture, Technology, etc.)
Sub-Category	Order Details	Product sub-category (Bookcases, Machines, etc.)
Quantity	Order Details	Number of units ordered
Month of Order Date	Sales Target	Month for which the sales target is defined
Target	Sales Target	Sales target amount for that month and category

Data Cleaning and Transformation

The following eight steps were carried out in Power Query Editor to clean, transform, and enrich the raw datasets:

1 Row Limiting and Data Types

Restricted the List of Orders table to the first 500 rows. Set the Order Date column to Date type and changed the Amount column to Fixed Decimal Number for accurate monetary calculations.

2 Text Formatting

Applied the Capitalize Each Word transformation to the CustomerName column to ensure consistent Proper Case formatting across all customer records.

3 Column Creation - Location

Merged the City and State columns into a new derived column called Location, formatted as City, State (for example: Mumbai, Maharashtra).

4 Column Creation - Profit Margin

Added a custom column with the formula: Profit Margin = Profit divided by Amount, expressed as a decimal to represent the profit percentage per order.

5 Conditional Column - Profit Status

Created a conditional column with three outcomes: values below zero are labelled Loss, values equal to zero are labelled Break-Even, and values above zero are labelled Profit.

6 Data Merging - Orders Data

Merged the List of Orders table with the Order Details table using Order ID as the join key (Left Outer Join). Expanded the resulting columns: Quantity, Category, and Sub-Category.

7 Duplicate and Missing Data Handling

Applied the Remove Duplicates step to the Orders Data table. Null values were retained in rows where Order IDs did not match between the two source tables.

8 Sorting and Filtering

Sorted the Orders Data table by Order Date in descending order, then applied a filter to retain only records from Tamil Nadu for regional analysis.

Grouping and Aggregating Data

Four aggregation tables were created by referencing base tables in Power Query Editor using the Group By operation.

Count by Order ID

Source: Order Details

Parameter	Value
Group By	Order ID
New Column	Count of Order ID
Operation	Count Rows

Average Profit by Category

Source: Order Details

Parameter	Value
Group By	Category
New Column	Average Profit
Operation	Average
Column	Profit

Total Amount by Sub-Category

Source: Order Details

Parameter	Value
Group By	Sub-Category
New Column	Total Amount
Operation	Sum
Column	Amount

Total Target by Month

Source: Sales Target

Parameter	Value
Group By	Month of Order Date
New Column	Total Target
Operation	Sum
Column	Target

Visualization and Insights

Descriptive Analysis

Total of 600 order records in Order Details across 4 categories.

List of Orders restricted to 500 rows with proper data types applied.

Average Profit: Furniture Rs. 1,258 | Office Supplies Rs. 1,430 | Technology Rs. 1,483 | Clothing Rs. 1,449.

Most orders were placed in Maharashtra, Tamil Nadu, and Karnataka.

Diagnostic Analysis

Row B-26118 in List of Orders has a null CustomerName value, which was retained with justification.

Order IDs in List of Orders (B-261xx) and Order Details (B-255xx) fall in different ranges. As a result, the Left Outer Join produces null values for Quantity, Category, and Sub-Category in the merged table.

Profit values below zero are identified as Loss in the Profit Status conditional column.

Duplicate rows were removed from the Orders Data table using the Remove Duplicates step.

Predictive Analysis

Technology shows the highest average profit at Rs. 1,483, indicating potential for increased stock investment.

The Tamil Nadu filter applied to Orders Data demonstrates a regional performance tracking capability.

Monthly target data in the Sales Target table allows forecast vs. actual comparison across time periods.

The Total Target by Month aggregation supports trend analysis across 96 monthly records.

Prescriptive Analysis

Increasing inventory for Technology and Clothing categories is recommended based on higher average profit margins.

The Order ID range mismatch between datasets should be resolved to enable accurate cross-table analysis.

Proper Case formatting should be applied consistently across all customer-facing data columns.

Data model relationships built on Order ID and Category can be used to create cross-table dashboards in Power BI.

Data Modeling - Relationships

Two active relationships were established in Power BI Model View using Manage Relationships:

Relationship 1 - Order ID

From Table	Column	To Table	Column	Cardinality
List of Orders	Order ID	Order Details	Order ID	Many to One (*:1)

Relationship 2 - Category

From Table	Column	To Table	Column	Cardinality
Order Details	Category	Sales Target	Category	Many to One (*:1)

Queries in Power Query Editor

A total of 8 tables/queries were present in the Power Query Editor on completion of the assignment:

Order Details	Sales Target	List of Orders	Orders Data
Count by Order ID	Avg Profit by Category	Total Amount by SubCategory	Total Target by Month

Conclusion

Data Import	Successfully imported 3 CSV datasets and applied all required Power Query transformations, including data types, text formatting, and calculated columns.
Data Merging	Created a merged Orders Data table using a Left Outer Join on Order ID, demonstrating the merge technique correctly even with non-overlapping ID ranges.
Aggregation	Built 4 aggregation tables using Group By operations: Count by Order ID, Avg Profit by Category, Total Amount by SubCategory, and Total Target by Month.
Sorting and Filtering	Applied sorting on Order Date (descending) and a state filter for Tamil Nadu to the Orders Data table for targeted regional analysis.
Data Modeling	Established 2 active data model relationships: List of Orders to Order Details on Order ID, and Order Details to Sales Target on Category.
Final Deliverable	Saved the completed workbook as PowerBI_Assignment1.pbix and documented all transformation steps for submission.